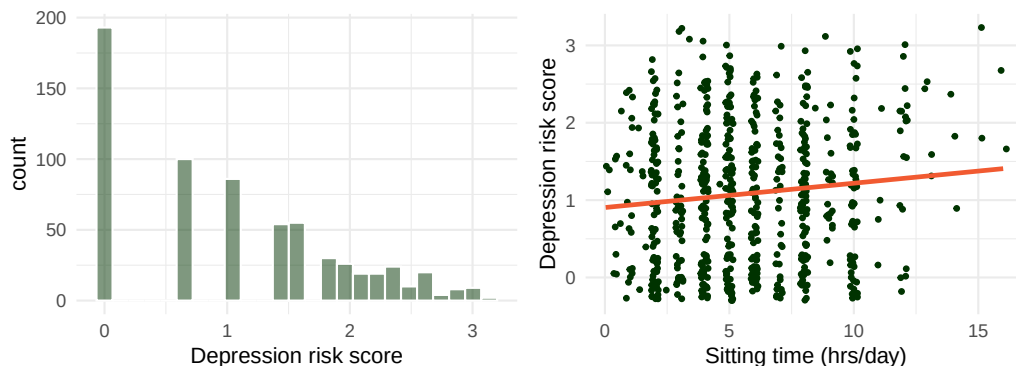


Day 02: Simple Linear Regression (SLR)

Data for today

The *National Health and Nutrition Examination Survey* (NHANES) is an ongoing CDC survey that collects health and behavioral data from a nationally representative sample of the U.S. population. Today we return to the sitting time and depression data from last class – this time looking at the actual numeric variables instead of the “High”/“Low” groups. We’ll summarize each participant’s responses to a 9-item depression screening survey as a **depression risk score** (higher = more risk).



The data also includes each participant’s age and sex. For our first analysis, we’ll investigate the relationship between **sitting time** and **depression symptoms**.

Regression, Intro Stat version

Q: describe the scatterplot

There is a _____ (strength), _____ (direction), _____ (form) association between sitting time and depression risk score.

Q: Write out the regression equation

```
lm(dep_score ~ sedentary_hours, data = nhanes)
```

Call:

```
lm(formula = dep_score ~ sedentary_hours, data = nhanes)
```

Coefficients:

(Intercept)	sedentary_hours
0.90426	0.03141

Q: Interpreting the slope:

i Note

The slope describes an *association* – an increase in sitting time doesn't *cause* an increase in depression symptoms just because the model says so. Watch for language that implies causation from an observational study.

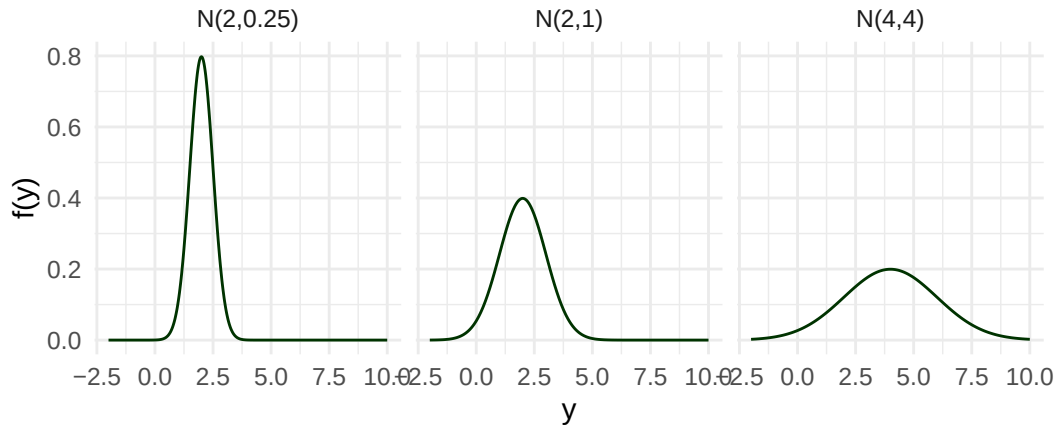
Q: make a prediction

A participant reports sitting for 6 hours per day. The model predicts:

Regression, Stat230 Version

Recap: the Normal distribution

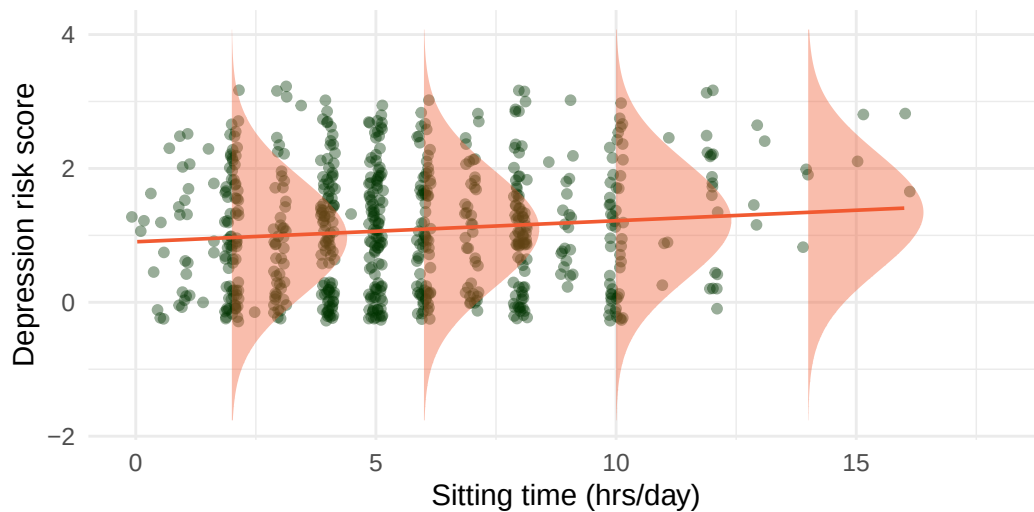
If Y is **numeric**, **unimodal**, and **symmetric**, we might use a Normal model: $Y \sim N(\mu, \sigma^2)$.



The Simple Linear Regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{where} \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

A **deterministic** model $Y_i = \beta_0 + \beta_1 x_i$ (no error term) would place every point exactly on the line – no uncertainty. Real data doesn't look like that. Instead, at every x value we imagine a full Normal *distribution* of possible y 's, centered on the line, all sharing the same spread σ :



SLR assumptions (mnemonic: think “LINE”):

- **L**inear relationship between x and y
- Errors are **I**ndependent and identically distributed
- Errors are **N**ormally distributed
- **E**qual variance of errors across all x (doesn't depend on x)
- (and errors have mean 0)

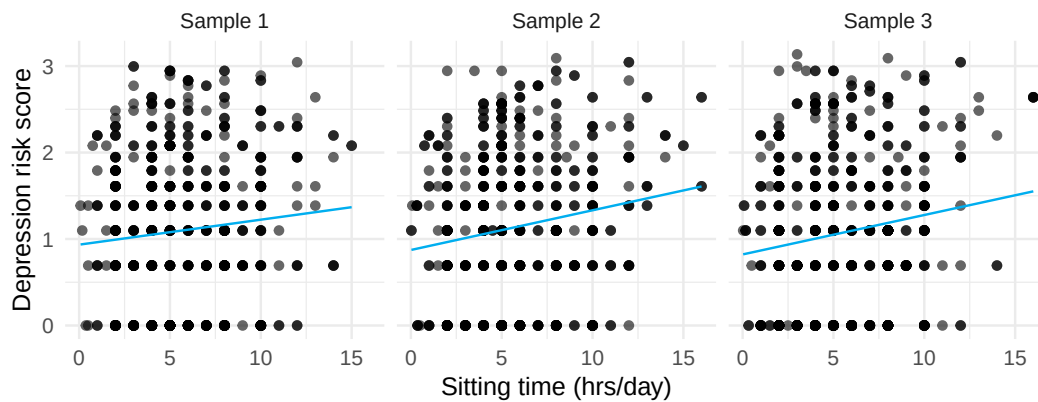
Notation

Mean function: $E(Y | X) = \mu\{Y | X\} = \beta_0 + \beta_1 x_i$

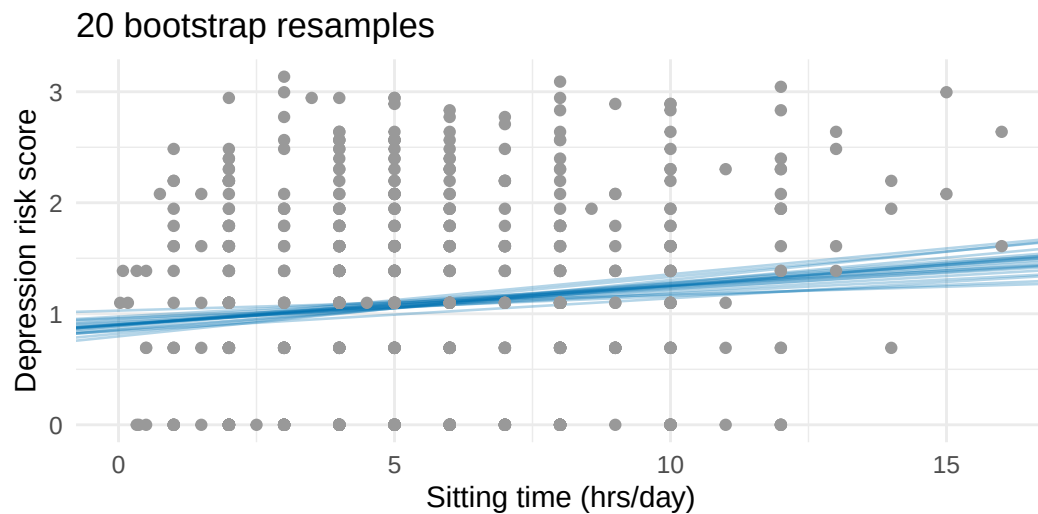
Fitted model equation: $\hat{Y}_i = \hat{\mu}\{Y | X\} = \hat{\beta}_0 + \hat{\beta}_1 x_i$

Uncertainty in the parameters

Every new sample of participants gives a slightly different fitted line. Below, each panel refits the model on a different bootstrap resample of the NHANES data:



Overlaying all 20 resampled lines on one plot shows that variability directly – that spread in the fitted _____ (not the raw data) is exactly what will let us build confidence intervals and run hypothesis tests next class:



Rest of today

 Group task

Open the SLR in R activity on Moodle and complete it in your groups. Make note of any questions, and choose someone responsible for reporting out your group's answers.